

Title

WHONDRS 2016 Sediment Organic Matter Characterization Data from Streams across HJ Andrews Experimental Forest, Oregon

Summary

This dataset supports a broader synoptic effort to map morphological, hydrological, chemical, and biological conditions across a fifth-order mountain stream network. Samples were generated through a collaborative synoptic sampling effort in 2016. The dataset provides sediment Fourier Transform Ion Cyclotron Resonance Mass Spectrometry (FTICR-MS) from 60 sites across the HJ Andrews Experimental Forest, Oregon (<https://andrewsforest.oregonstate.edu>). Related data were collected as part of the event and were published separately in collaboration with other team members. The data are available at <http://www.hydroshare.org/resource/ea6c0832885a46c3939e7bb22e48e754> and are described within <https://doi.org/10.5194/essd-11-1567-2019> (Ward et al., 2019). The hydroshare data package contains processed FTICR-MS data from the samples included in this data package. The data were processed via Formultitude (previously called Formularity; <https://github.com/PNNL-Comp-Mass-Spec/Formultitude>). However, we have re-processed the data using Core-MS and included it in this data package. Additional related data collected in 2025 from a similar effort can be found at <https://data.ess-dive.lbl.gov/datasets/doi:10.15485/3023310> and <http://www.hydroshare.org/resource/b274c4a234bf4b12b7cb8a54a696c629>.

For details on how to navigate data packages generated by this project, see <https://data.ess-dive.lbl.gov/portals/PNNLRiverCorridorSFA/About>.

Brief Overview of Methods

Samples were collected on July 26 – August 3, 2016 from 60 sites across HJ Andrews Experimental Forest, Oregon, United States via collaboration between WHONDRS (<https://whondrs.pnnl.gov>) and the Synoptic Sampling Team Members. Sediment samples were collected underwater at 1 to 3 centimeters depth below the streambed and sieved to less than 2 millimeters. The sediment samples were stored on dry ice upon collection and shipped with dry ice to Pacific Northwest National Laboratory.

Prior to Fourier transform ion cyclotron resonance mass spectrometry (FT-ICR-MS) analysis, organic matter was extracted from sediments by adding 1 mL of ultrapure water (18 MΩ) to 500 mg of sediment. Each sample was extracted three times, the supernatants were combined and diluted to 5 mL, acidified to pH 2 with 85% phosphoric acid, and concentrated using Bond Elut Priority PolLutant (PPL) solid-phase extraction cartridges. Samples were analyzed on a 12 Tesla (12 T) Bruker solariX FT-ICR-MS using electrospray ionization (ESI) in negative ion mode. Samples were infused at 3.0 $\mu\text{L min}^{-1}$, with an ion accumulation time of 0.1 seconds, a needle voltage of +4.4 kV, and a capillary temperature of 180 °C. Data were acquired over a mass-to-charge (m/z) range of 98–900 with 144 co-added scans per sample. Unprocessed and processed data are provided to allow the user flexibility in data processing. Instructions and scripts to process the data using CoreMS are provided. For details regarding laboratory and instrument methods, see the associated manuscript (<https://doi.org/10.5194/essd-11-1567-2019>).

Critical Details

- 1 – **Sample and file naming:** The samples and files within this data package follow a consistent naming scheme to enable easy use of data and support AI-readiness. The sample names follow the format

“HJA_[Site_ID]_p1”, where “p1” indicates an ion accumulation time of 0.1. The field metadata contains the site ID and can be mapped across each other accordingly. All files begin with “WHONDRS”, a three-letter code that indicates the study – “HJA”, and the year of the study “2016”. The text following these indicators and separated with an underscore describes the material and data type(s) within the file.

- 2 – **FTICR-MS processing levels:** FTICR-MS data is provided in three processing levels in this data package. We have included pre-processed (i.e., peak picked) XML files, CoreMS output files, and fully processed data to allow for maximum data user flexibility. The unprocessed instrument files are available from the Environmental Molecular Sciences Laboratory (EMSL). These can be processed in newly developed public workflows (i.e., CoreMS) for data use or they can be first processed in a proprietary software (i.e., Bruker) into XML files and then processed in publicly available Formultitude (previously called Formularity; <https://github.com/PNNL-Comp-Mass-Spec/Formultitude>) for further data use. In previous data packages, we have processed the files in Bruker prior to publishing and then only published XML files and instructions for the user to process the XML files in Formultitude.
- 3 – **FTICR-MS processing instructions:** We have provided a static version of our CoreMS processing instructions and scripts that were used for this dataset. These instructions and scripts may change over time as this new pipeline evolves. The most up to date instructions and scripts can be found at https://github.com/danczakre/RCSFA_CoreMS/tree/main/. A release was created on GitHub to document the version pulled for this data package.
- 4 – **FTICR-MS data synthesis:** We do not recommend combining **processed** FTICR-MS data, whether it was run through Formultitude or CoreMS (*CoreMS_Processed_ICR_*.csv). If you wish to combine data from across datasets, you need to go back to at least the .xml files and reprocess the data using your pipeline of choice. For convenience, we have included the CoreMS outputs (*.corems.csv files) which **can** be combined with other CoreMS outputs released in our data packages, though we still recommend reprocessing with your pipeline. We are happy to work through these considerations with you if you have questions.
- 5 – **Coordinates:** The coordinates provided are approximately where samples were taken. They are not exact.
- 6 – **Legacy Data:** This data package contains legacy data from 2016 and therefore does not contain all of the standardized metadata and formats that are typically found in other WHONDRS data packages.

Data Package Structure

In addition to this readme, this data package also includes a file-level metadata (FLMD) file that describes each file and a data dictionary (DD) that describes all column/row headers and variable definitions.

This dataset is comprised of (1) a folder of sample data; (2) data dictionary; (3) file-level metadata; (4); (5) coordinates; and (6) readme. The sample data subfolder contains 12 Tesla (12T) FTICR-MS data. This folder contains the processed data and three subfolders, one containing the .xml files, one containing the CoreMS output files, and the other containing instructions and scripts for processing the files in CoreMS (<https://github.com/EMSL-Computing/CoreMS>). All files are .csv, .pdf, .R, .xml, .Rmd, .py, .cal, or .json.

Citations, Acknowledgements, and License

Acknowledgments

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Citations:

- Ward, A. S., Zarnetske, J. P., Baranov, V., Blaen, P. J., Brekenfeld, N., Chu, R., Derelle, R., Drummond, J., Fleckenstein, J. H., Garayburu-Caruso, V., Graham, E., Hannah, D., Harman, C. J., Herzog, S., Hixson, J., Knapp, J. L. A., Krause, S., Kurz, M. J., Lewandowski, J., Li, A., Martí, E., Miller, M., Milner, A. M., Neil, K., Orsini, L., Packman, A. I., Plont, S., Renteria, L., Roche, K., Royer, T., Schmadel, N. M., Segura, C., Stegen, J., Toyoda, J., Hager, J., Wisnoski, N. I., and Wondzell, S. M.: Co-located contemporaneous mapping of morphological, hydrological, chemical, and biological conditions in a 5th-order mountain stream network, Oregon, USA, *Earth Syst. Sci. Data*, 11, 1567–1581, <https://doi.org/10.5194/essd-11-1567-2019>, 2019.
- Ward, A. (2024). River Corridor - ESSD - Data Collection (2024 Update), HydroShare, <http://www.hydroshare.org/resource/ea6c0832885a46c3939e7bb22e48e754>
- Ward, A. S. (2026). River Corridor Post-Fire Synoptic - ESSD, HydroShare, <http://www.hydroshare.org/resource/b274c4a234bf4b12b7cb8a54a696c629>
- Garayburu-Caruso V A ; Danczak R E ; Forbes B ; Gillespie X ; Goldman A E ; Renteria L ; River Corridor Critical Zone Workshop Attendees T ; Stegen J C (2026): WHONDRS Surface Water and Sediment Geochemistry and Organic Matter Characterization Data from Streams across HJ Andrews Experimental Forest, Oregon. River Corridor Hydro-biogeochemistry from Molecular to Multi-Basin Scales SFA, ESS-DIVE repository. Dataset. [doi:10.15485/3023310](https://doi.org/10.15485/3023310)

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Change History

Change history:

Data Package Version	Changes
Version 1 <i>July 2026</i>	Original data package publication